



Homeostasis and response

Name: _____

Class: _____

Date: _____

Time:

Marks: **64 marks**

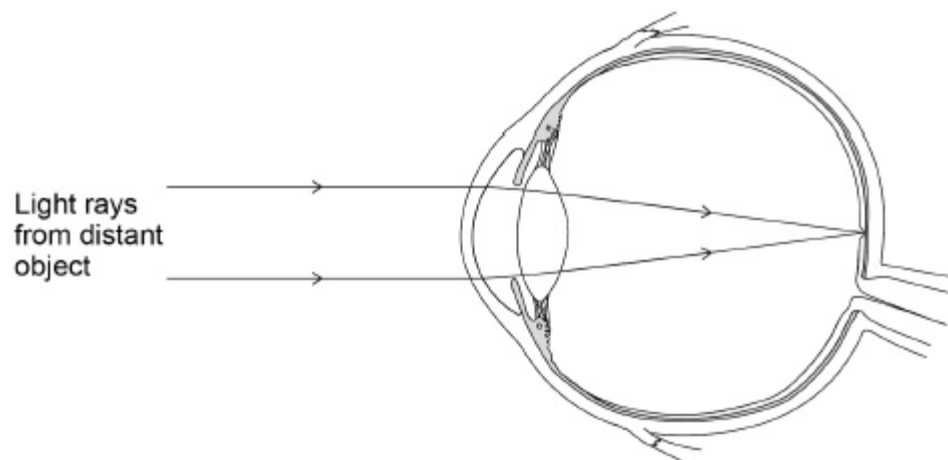
Comments:

1.

The human eye can focus on objects at different distances.

Figure 1 shows how a clear image of a **distant** object is formed in a person's eye.

Figure 1



(a) Explain how the person's eye could adjust to form a clear image of a **nearer** object.

(6)

- (b) Explain why a long-sighted person has difficulty seeing near objects clearly.

(2)

- (c) Long-sightedness can be corrected by wearing spectacles.

Describe how spectacle lenses can correct long-sightedness.

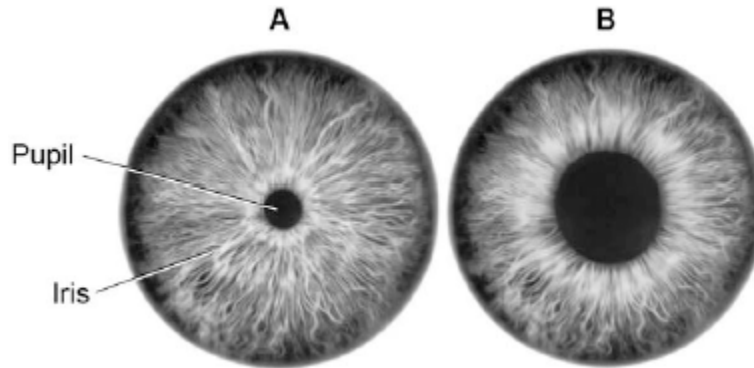
(3)

(Total 11 marks)

2.

Figure 1 shows a reflex in the iris of the human eye in response to changes in light levels.

Figure 1



@ Gande Vasas/Stone/Getty Images

- (a) Describe the changes in the pupil and iris going from **A** to **B** in **Figure 1**.

Explain how these changes occur.

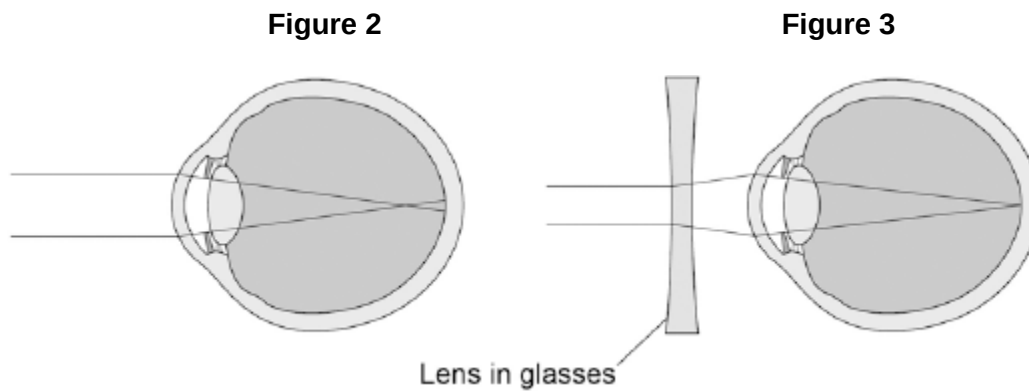
Refer to the changes in light level in your answer.

(4)

- (b) Some people wear glasses to improve their vision.

Figure 2 shows light entering the eye in a person with blurred vision.

Figure 3 shows how this condition is corrected with glasses.



Compare **Figure 2** and **Figure 3**.

Explain how the blurred vision is corrected.

(2)
(Total 6 marks)

3.

It is important to keep the blood glucose concentration within narrow limits.

- (a) A person eats a meal containing a lot of carbohydrate. This causes an increase in the person's blood glucose concentration.

Explain how the hormones insulin **and** glucagon control the person's blood glucose concentration after the meal.

(5)

- (b) The body cells of a person with Type 2 diabetes do **not** respond to **insulin**.

A person with Type 2 diabetes often has a higher blood **insulin** concentration than a non-diabetic person.

Explain why.

(3)

Metformin is a drug used for treating people who have Type 2 diabetes.

Scientists investigated the effects of metformin and two other drugs, **A** and **B**.

The scientists wanted to see how the drugs affected the blood glucose concentrations of 220 people with Type 2 diabetes.

This is the method used.

1. Put the 220 people into five groups.
2. Treat each group with a different drug or combination of drugs for several weeks.
3. Give each person a meal high in carbohydrate.
4. Measure the blood glucose concentration of each person 30 minutes after the meal and again 3 hours after the meal.

(c) Suggest **three** variables that the scientists should have controlled in the investigation.

- 1 _____

- 2 _____

- 3 _____

(3)

The scientists recorded their results as a mean value for each group.

The scientists calculated the 'standard deviation' for each group's result.

Standard deviation is a measure of the spread of the individual results above or below ($\pm$) the mean value.

The scientists gave each group's result as:

mean $\pm$ standard deviation

The larger the standard deviation, the greater is the spread of results around the mean.

(d) Which of the results is the most precise?

Tick ($\checkmark$) **one** box.

Mean = 171.6 ± 16.3

☐

Mean = 177.2 ± 15.4

☐

Mean = 182.5 ± 18.2

☐

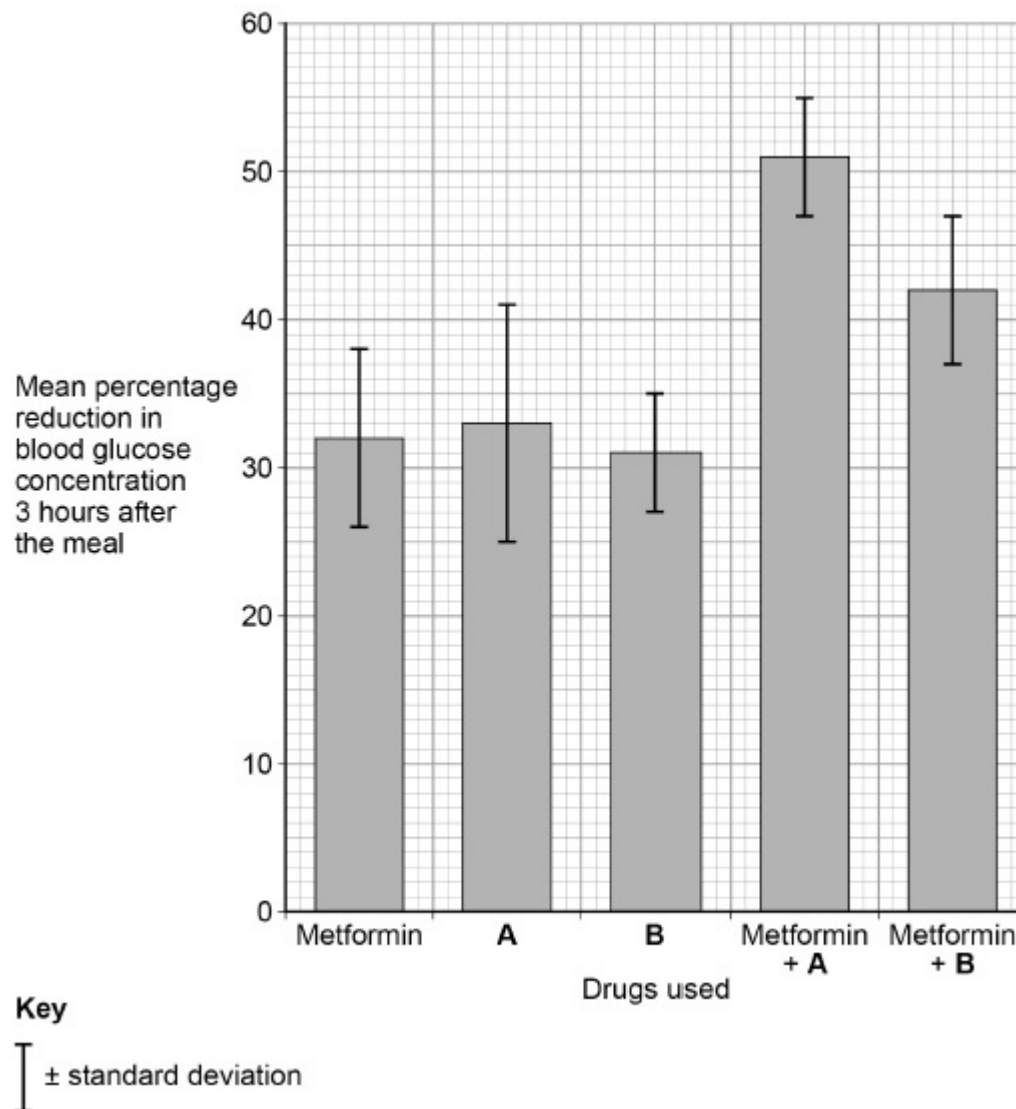
Mean = 205.2 ± 19.4

☐

(1)

The following table and the figure show the scientists' results.

Drugs used	Metformin	A	B	Metformin + A	Metformin + B
Number of people	60	40	25	65	30
Mean blood glucose concentration 30 minutes after the meal in mg/100 cm ³ $\pm$ standard deviation	177.2 $\pm$ 15.4	182.5 $\pm$ 18.2	171.6 $\pm$ 16.3	205.2 $\pm$ 19.4	206.5 $\pm$ 19.6



In the table and the figure some standard deviations of results overlap.

- An overlap of standard deviations shows the difference between the means is **not** significant.
 - **No** overlap of standard deviations shows a significant difference between the means.
- (e) A student looked at the scientists' method and the results in the table and figure above.

The student stated:

'Metformin works better when used with other drugs.'

Evaluate the student's statement.

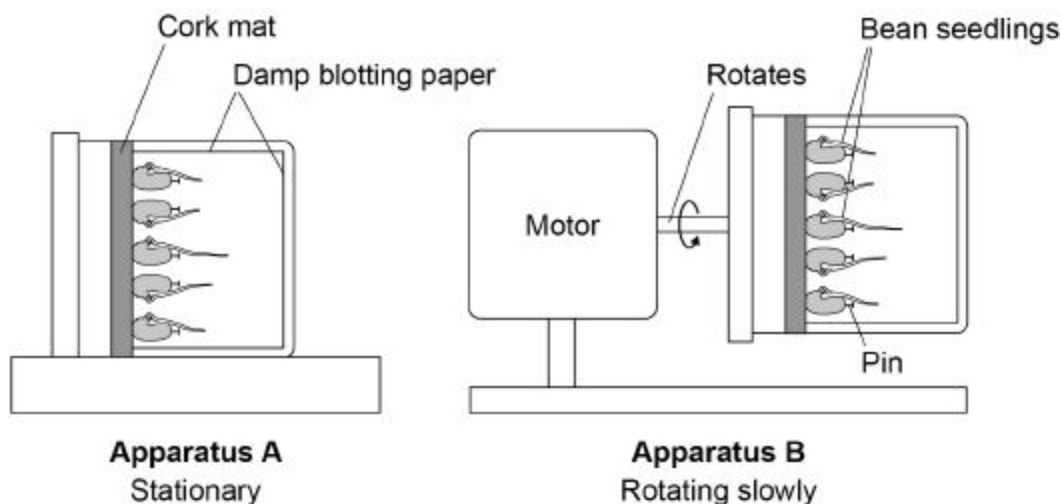
(6)
(Total 18 marks)

4.

Some students investigated geotropism in the roots of bean seedlings.

Figure 1 shows the apparatus used.

Figure 1



This is the method used.

1. Measure the length of the root of each of 10 bean seedlings.
 2. Pin 5 seedlings to the cork mat in apparatus **A**.
 3. Pin 5 seedlings to the cork mat in apparatus **B**.
 4. Leave **A** and **B** in a dark cupboard for 2 days.
 5. After the 2 days:
 - make a drawing to show the appearance of each seedling
 - measure the length of the root of each seedling.
- (a) Why did the students surround the seedlings with damp blotting paper?

Tick **one** box.

To prevent light affecting the direction of root growth

☐

To prevent photosynthesis taking place in the roots

☐

To prevent the growth of mould on the roots

☐

To prevent water affecting the direction of root growth

☐

Apparatus **B** is a control.

Apparatus **B** rotates slowly.

(b) How does apparatus **B** act as a control?

(1)

The table below shows the students' results.

	Apparatus A					Apparatus B				
Seedling number	1	2	3	4	5	1	2	3	4	5
Length at start in mm	35	41	32	33	39	30	33	29	28	31
Length after 2 days in mm	49	57	43	45	54	45	45	44	29	44
Length change in mm	14	16	11	12	15	15	12	15	1	13
Mean length change in mm	14					11				

(c) One student stated:

'The mean length change for the seedlings in apparatus **B** is **not** valid.'

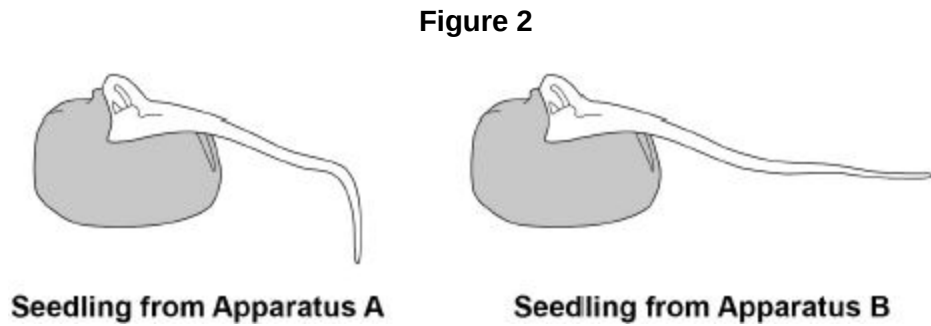
Suggest the reason for the student's statement.

(1)

- (d) Suggest **one** improvement the students could make to obtain a more valid mean length change for the seedlings in apparatus **B**.

(1)

- (e) **Figure 2** shows the students' drawings of two seedlings at the end of the 2 days.



A plant hormone is made in the root tip.

The hormone diffuses from the tip into the tissues of the root.

Explain how the hormone causes the appearance of the seedlings in **Figure 2** to be different.

You should refer to **both** seedlings in your answer.

(3)

- (f) In horticulture plant hormones are used for controlling plant growth.

Draw **one** line from each plant hormone to the correct use of that hormone.

Plant hormone	Use of hormone
Auxin	To reduce the time taken for tomatoes to ripen
Ethene	To slow down the growth of plant stems
Gibberellin	To promote seed germination
	To stimulate root growth in plan cuttings

(3)

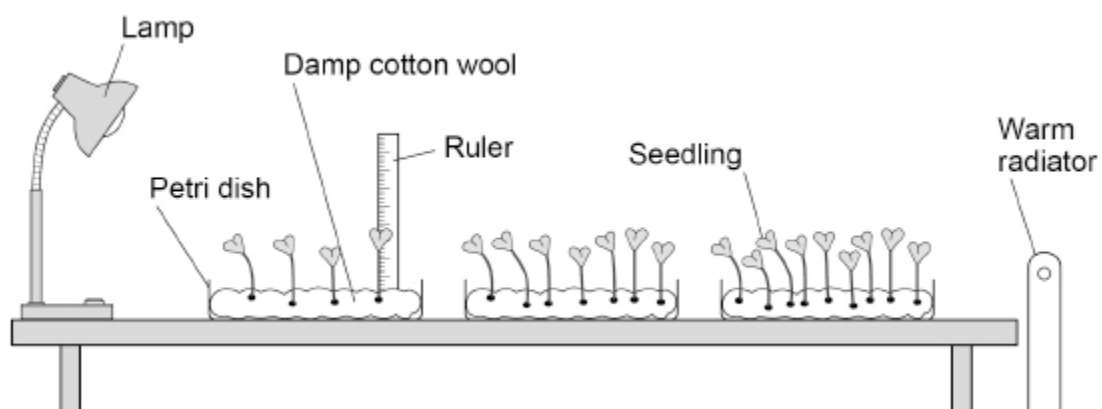
(Total 10 marks)

5.

A student investigated the effect of light intensity on the growth of seedlings.

Figure 1 shows the equipment.

Figure 1



(a) Which **two** improvements should the student make to the investigation?

Tick (✓) **two** boxes.

Give more water to the seedlings nearest the lamp

☐

Leave some of the seedlings for a few more days

☐

Open a window to let more air in

☐

Put all the dishes the same distance from the radiator

☐

Use equal numbers of seedlings in each dish

☐

(2)

(b) What is the dependent variable in the investigation?

Tick (✓) **one** box.

The height of the seedlings

☐

The mass of cotton wool

☐

The temperature of the room

☐

(1)

- (c) In each dish the seedlings compete with each other.

Give **two** factors the seedlings compete for.

1 _____

2 _____

(2)

Figure 2 shows a seedling growing towards a lamp.

Figure 2



- (d) What happened to the growth of the seedling on side **P** compared with the growth on side **Q**?

Tick (✓) **one** box.

Side **P** has grown less than side **Q**

☐

Side **P** has grown more than side **Q**

☐

Side **P** has grown the same as side **Q**

☐

(1)

- (e) Plant responses are called tropisms.

Which tropism causes the seedling to grow towards light?

Tick (✓) **one** box.

Geotropism

☐

Gravitropism

☐

Phototropism

☐

(1)

- (f) Which hormone causes the seedling to grow towards the light?

Tick (✓) **one** box.

Auxin

☐

Insulin

☐

Testosterone

☐

(1)

(Total 8 marks)

6.

Phototropism is a growth response by part of a plant to light.

- (a) Name **one** other tropism.

Give the stimulus the plant responds to in the tropism you have named.

Tropism _____

Stimulus _____

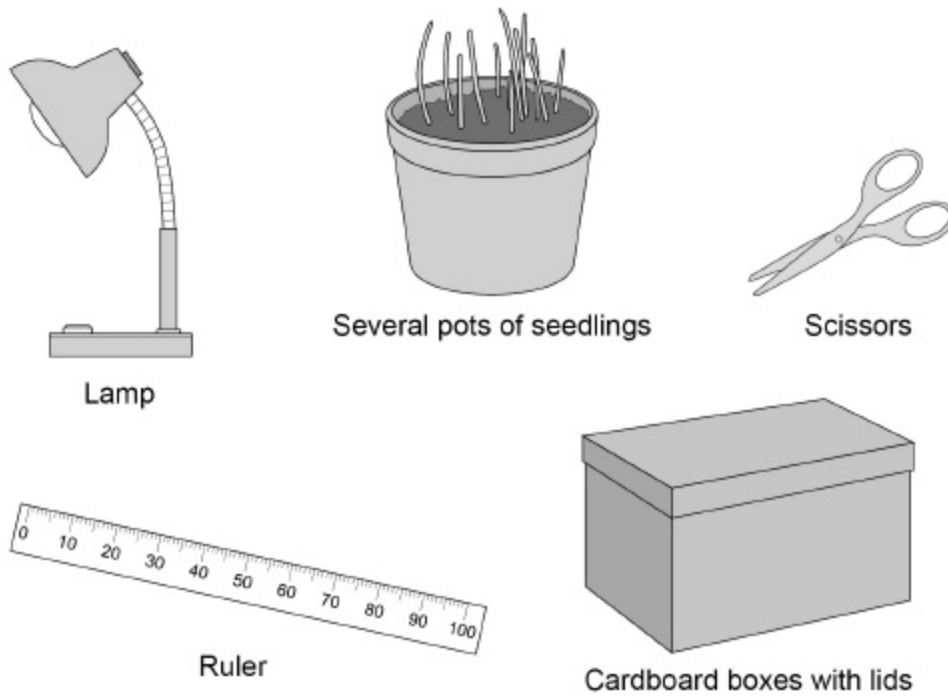
(2)

- (b) Plan an investigation to show the effect of light from one direction on the growth of plant seedlings.

Include details of any controls needed.

You may use some of the equipment shown in **Figure 1** and any other laboratory apparatus.

Figure 1



(6)

- (c) Explain how phototropism in a plant shoot helps the plant to survive.

(3)

(Total 11 marks)

Mark schemes

1.

- (a) ciliary muscles contract

1

(so ciliary muscles have a) smaller diameter

1

(so) suspensory ligaments loosen / slacken

*do **not** accept 'relax'*

1

(so) lens thickens **or** lens becomes more curved / rounded

allow lens becomes fatter

ignore lens becomes bigger

1

(thicker) lens is more convergent

*allow light rays bent (inwards) more **or** light refracted more*

1

light rays / image focused on retina

allow light rays meet on retina

1

- (b) eye(-ball) is (too) short **or** lens cannot be thickened enough

*allow ciliary muscles (too) weak **or** lens not (sufficiently) elastic*

1

(so) light 'focuses' behind retina

allow (so) image forms behind retina

1

- (c) convex / converging lens

allow shape described eg thicker in middle

1

light rays bent / refracted (inwards) more

allow changes direction of light rays further inwards

1

light rays focused on retina

*allow light rays brought to a point on retina **or** light rays converge on retina **or** focused / clear image forms on retina*

1

[11]

2.

- (a) pupils dilated (at **B**)

*allow converse for **A***

1

in dim light / low light levels

1

because circular muscles (in iris) relax

1

(and) radial muscles contract

1

- (b) figure 2 shows myopia where light does not focus on the retina
allow refraction

1

in figure 3 the lens bends the light so that light focuses on the retina

1

[6]

3.

(a)

ignore incorrect organ secreting insulin / glucagon

(blood glucose increases after meal causing) insulin secretion
*allow (blood glucose increases after meal causing)
insulin increase*

1

insulin causes glucose to enter cells / liver / muscles

1

(insulin causes) glucose conversion to glycogen

1

*allow glucose converted to glycogen in cells / liver /
muscles for 2 marks*

(so) blood glucose decreases causing glucagon secretion
allow increase in glucagon when blood glucose is low

1

glucagon causes glycogen to be converted to glucose

1

- (b) cells / liver / muscles absorb less glucose
*allow cells / liver / muscles convert less glucose to
glycogen
do **not** accept no absorption / conversion of glucose*

1

(so) glucose concentration in blood remains high
*allow (so) glucose concentration in blood does not
decrease*

1

(high blood glucose stimulates / causes) pancreas to release more insulin
*allow more insulin is released from pancreas to 'try' to
reduce blood glucose*

1

- (c) any **three** from:
- age
 - height **and** mass
allow BMI
 - proportion of males and females **or** group size
allow sex of the participants
 - (same) severity of diabetes
 - (same) activity (during investigation)
 - (same) type of meal
 - dose of drug
 - (similar) blood glucose concentrations at start
allow how much / type of food / drink consumed before
 - other health conditions **or** other drugs being taken
allow may not have followed drug-taking regime beforehand

3

- (d) Mean = $177.2 + 15.4$

1

- (e) **Level 3:** A judgement, strongly linked and logically supported by a sufficient range of correct reasons, is given.

5–6

Level 2: Some logically linked reasons are given. There may also be a simple judgement.

3–4

Level 1: Relevant points are made. They are not logically linked.

1–2

No relevant content

0

Indicative content

Pro:

- Met + A gives larger (%) reduction (in blood glucose) than Met alone
- so statement is supported
- Met + B gives larger (%) reduction (in blood glucose) than Met alone
- so statement is supported
- Met + A SD does not overlap with Met SD
- so difference is significant

Con:

- Met + B SD overlaps with Met SD
- so difference is not significant
- difference in results could be due to chance

-
- number of people used is not very large
 - number of people in each group is different
 - so may not be representative **or** may not be repeatable / reproducible
 - so anomalies will have a bigger impact on smaller groups
 - 30 minute / starting levels of blood glucose are different
 - all 30 minute / starting levels are higher in the 2-drug trial
 - so may cause different % reductions
 - no information about control variables **or** named e.g.
 - concentration of drugs not given / may differ
 - so results may not be valid

for level 3 an inclusion of a discussion of significance is required

[18]

4.

- (a) to prevent water affecting the direction of root growth

1

- (b) gravity acts evenly on all sides

allow cancel out the effect of gravity

*do **not** accept there is no gravity*

1

- (c) (mean) includes the (anomalous) result for seedling 4

allow (mean) includes the (anomalous) result which only grew 1 mm

1

- (d) calculate (mean) from just seedlings 1, 2, 3 and 5

or

repeat the investigation **and** recalculate (a new mean)

allow omit seedling 4 from (mean) calculation

1

- (e) uneven distribution of hormone in (root / seedling of) A

allow reference to auxin

allow more hormone at bottom

*do **not** accept more hormone at the top*

1

even distribution of hormone in B

allow B does not have an uneven distribution of hormone

1

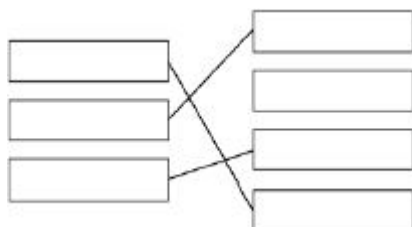
(so) top grows fast(er) (than bottom) in (root / seedling of) A (and equal growth in B)

allow (more) cell elongation or cell division on top of A

allow converse for lower surface

1

(f)



extra line for a hormone cancels mark for that hormone

1

1

1

[10]

5.

- (a) put all the dishes the same distance from the radiator

1

use equal numbers of seedlings in each dish

1

- (b) the height of the seedlings

1

- (c) any **two** from:

- light
- water
- mineral(s) / ions / salts

allow nitrate / magnesium / nitrogen / nutrients

allow space

ignore food

ignore carbon dioxide / oxygen

ignore heat

2

- (d) side **P** has grown less than side **Q**

1

- (e) phototropism

1

(f) auxin

1

[8]

6.

- (a) named example of tropism – e.g. geotropism / gravitropism
allow hydrotropism or chemotropism or thermotropism

1

correct corresponding stimulus – e.g. gravity

allow water or chemical or 'heat'

1

- (b) **Level 3:** The method would lead to the production of a valid outcome. All key steps are identified and logically sequenced.

5–6

Level 2: The method would not necessarily lead to a valid outcome. Most steps are identified, but the plan is not fully logically sequenced.

3–4

Level 1: The method would not lead to a valid outcome. Some relevant steps are identified, but links are not made clear.

1–2

No relevant content

0

Indicative content

- several seedlings in each batch **or** one pot of seedlings in each batch
- measure heights of shoots
- leave some in dark with light from one side / direction in box with hole
- control(s) with all-round light **or** rotating on clinostat **or** in dark
- control variable(s) e.g. same temperature / water / soil type
- after suitable time (at least several hours)
- record appearance of seedlings re. light direction
- re-measure heights of shoots
- detail of how bent shoots were measured – e.g. use thread or straighten them out
- calculate mean height increase for each group
- use ruler / protractor to estimate angle of bending

for **level 3** a reference to comparing the growth of plants with light from one direction with plants either in darkness or in full light along with a control variable is required

(c) leaves / plant receive(s) / absorb(s) more light

1

(so) more photosynthesis

1

(so plant) produces more glucose

*allow starch / carbohydrate / sugar / organic material /
other named organic substance*

*if no other mark awarded allow 1 mark for any two of the
mark points with no reference to 'more'*

1

[11]

Examiner reports

1.

- (a) This question differentiated between students very well. Given an unlabelled diagram of the formation of an image of a distant object by the eye (Figure 1), students had to describe how the eye would adjust to form an image of a nearer object. Although many knew that the ciliary muscles would contract (rather than 'constrict'/'tighten'), some had forgotten the name of the muscles and others thought the muscles would relax. Only a small minority realised that this would give a smaller diameter to the circle of muscle tissue.

A very common error was to state that the suspensory ligaments would then 'relax' rather than slackening or loosening. Most knew that the lens would then bulge or become 'fatter', that it would become more convergent (or refract the light rays more, but not 'more light is refracted' and certainly not 'light is reflected more') and the image would then be in focus on the retina. The idea of light simply 'hitting' the retina was not creditworthy.

- (b) Half of the students scored any marks on this question. Around 13% of students understood completely that a person who is long-sighted cannot form a clear image of a near object either because the eyeball is too short or the lens cannot be thickened sufficiently and so the image would have been 'focused' behind the retina.

Many students were clearly confused about the nature of long-sightedness and short-sightedness and told the wrong story. Many others confused accommodation with adjustment to a change in light intensity and wrote about the action of the iris.

- (c) Many students had difficulty explaining how spectacles could correct long-sightedness. Some were confused over whether a converging or diverging lens was required (usually described, respectively, as convex or concave), although most knew that the light rays from the near object would need to be refracted (inwards) more. Many stated that the light from the near object could then be brought to a focus 'on the back of the eye' rather than specifying the 'retina' and did not gain credit.

This question differentiated very well across the ability range with around 12% of students scoring full marks.

3.

- (a) About 31% of students were able to give a complete account of how insulin and glucagon would control a person's blood glucose concentration after a meal. These students knew that insulin would be released to cause glucose to enter the body cells, that the glucose would be converted into glycogen, that the consequential lowering of the blood glucose level would cause glucagon secretion and that this stimulated the conversion of glycogen back to glucose.

Some students confused the roles of the two hormones and often had problems making themselves understood due to hybrid spellings between the terms glucagon and glycogen. Despite glucose being mentioned in the question three times, many students wrote about 'blood sugar'.

- (b) This question differentiated well between the higher-attaining students. Overall, around 36% scored marks with a very small proportion of this gaining all three. Since the cells of a person with Type 2 diabetes do not respond to insulin, this would result in less glucose entering the cells (not 'no glucose' entering as many students claimed, which would have resulted in the death of the cells). Consequently, much glucose would remain in the blood, raising the blood's glucose concentration which in turn would stimulate the pancreas to keep on releasing insulin, hence the elevated insulin levels.

Answers that did not gain credit often referred to 'amounts' of glucose 'in the body' rather than concentrations of glucose in the blood and the notion that, since insulin was not 'used up' in stimulating the cells, it would therefore remain in the blood leading to the observed high insulin concentration.

- (c) Nearly all students were able to suggest at least one control variable for the investigation, such as:

- the age of the participants
- their gender
- BMI
- severity of diabetes
- the dose of the drug given
- the same meal given to each
- other health conditions
- other drugs being taken.

A third of students gave the required three examples.

- (d) The stem of this question introduced the students to the unfamiliar concept of standard deviation. No prior knowledge of this concept was required as sufficient detail was given to students to allow them to make use of it in answering parts (d) and (e).

Given that standard deviation was a measure of the spread of results about the mean. Around 72% of students had no problem in selecting the answer 'mean = 177.2 ± 15.4 ' as being the most precise from the four examples given as this had the smallest standard deviation.

- (e) This question was intended for the highest-attaining students and differentiated well at this level. A table of data and a bar graph of more data had to be used in the evaluation of a student's statement that the drug metformin worked better when used with other drugs. Students had to decide what was meant by 'worked better' and they had to select appropriate data to use in their evaluation as not all of the data was relevant.

The question was marked using a level of response mark scheme, with three levels. The descriptors for Level 2 (3–4 marks) and Level 3 (5–6 marks) required that 'some logically linked reasons are given' and, since this was an evaluation, at least one logically linked reason supporting the statement and at least one that did not support the statement were required. Many students did not meet this criterion, and around 84% of students were limited to Level 1 or below.

Both the table and the graph gave the standard deviation (SD) of each result. The bullet points immediately before the actual question explained that an overlap of SDs between two means shows there is no significant difference, whereas non-overlap can be taken to show a significant difference.

Some students concentrated too much on the data in the table and claimed that the blood glucose concentrations in people in the 2-drug trial were much higher than those in the 1-drug trial. Although this was true, it was a result after only 30 minutes (the graph showed results after 3 hours) and all the SDs overlapped, indicating that the differences were insignificant. Some of these students also thought that the higher blood glucose level was an 'improvement' of the patients' condition.

The graph, however, showed clearly that the mean reduction in blood glucose concentration was higher when metformin was used with drug A *and* when used with drug B, which supported the student's statement. In addition, the SD for metformin + drug A did not overlap with the SD for metformin alone and hence showed a *significant* improvement, thus supporting the student's statement. However, the results for metformin + drug B showed a small overlap of SDs with metformin on its own and hence showed *insignificant* improvement. Consideration of the effects of drug A and drug B on their own were irrelevant (although many students included these in their answers).

Most students did not give any consideration to the scientists' methodology. However, some mentioned the low numbers of people in each drug trial and some mentioned that there were different numbers in each trial, but few developed this further in terms of anomalies having a greater effect on smaller groups. Hardly any students stated that there was no mention of other control variables in the scientists' investigation, despite having themselves suggested such control variables in their answer to part (c); had these not been controlled, it might have invalidated the results.

Finally, to attain Level 3, a judgement had to be made on whether the student's statement was correct or incorrect or partially correct. This judgement had to be supported by a discussion of the evidence, as presented above.

4.

- (a) In the stem of the question students were told that both sets of apparatus were placed in a dark cupboard. However, many students thought that the purpose of the damp blotting paper in each set of apparatus was to prevent light affecting the direction of root growth.

57% of students selected the correct reason which was to prevent water affecting the direction of root growth.

- (b) 24% of students were able to explain that apparatus B acted as a control because it allowed gravity to act evenly on all sides or cancelled the effect of gravity.

Many students attempted to explain why a control was needed rather than how apparatus B acted as a control in the given situation. Many also misused the terms 'geotropism' or 'gravitropism' when they in fact meant gravity.

- (c) Lack of precision was an issue for many students here: while most realised there was an anomalous result for apparatus B, many did not point out that this was for seedling 4 whose increase in length was less than 10% of that of the other seedlings.
- (d) Two omissions were evident in the answers of most students. As in question 06.3, many forgot to mention seedling 4. Plus, although most thought the anomalous result should have been ignored or that the experiment should have been repeated, they did not state that the mean should have been recalculated. 31% of students achieved the mark.

- (e) Although the drawing of the root of a seedling from apparatus A showed a right-angle bend with the tip pointing vertically downwards, many students stated that the effect was due to a response to light.

34% of students gained partial credit and 13% achieved all three marks for stating that gravity had caused the hormone to be unevenly distributed in the root of seedling A resulting in more growth on the upper side. While in seedling B the hormone was evenly distributed.

- (f) 61% of students were able to link each of the plant hormones, auxin, ethene and gibberellin, to a correct example of its use in horticulture.

5.

In (b), fewer than half of the students correctly identified the dependent variable. Of the majority giving an incorrect response, the most common, by far, was the temperature of the room. For (c) most students gave at least one correct response, usually identifying light or water as factors being competed for; although many students gave both of these. The other most commonly suggested allowable response was 'space'. In (d), a large proportion of students gave the incorrect answer, that side P (facing the light) grows more than side Q – an obvious misconception about the effect of more or faster growth on one side of the shoot. Given the clear bend in the shoot, a large number suggested that the two sides had grown equally.

6.

- (a) Having been given a definition of phototropism, less than one-third of students were able to name a second tropism. The most common correct answer was 'gravitropism' or 'geotropism', although 'hydrotropism' was given by a few. Two-thirds of these students were able to name the corresponding stimulus. Some students described a response rather than just giving the stimulus.
- (b) In this question, students had to design an investigation to show the effect of light from one direction on the growth of plant seedlings. Figure 1 showed equipment that could be used in the investigation.

This extended response question was marked on the basis of a 'level of response' mark scheme. An answer at Level 2 (3–4 marks) had a minimum requirement of seedlings being set up in different light conditions so that the effect of these could be compared. Level 3 requires students to describe an investigation that would give valid results and therefore a Level 3 answer (5–6 marks) required comparison of the effect of light from one direction (in terms of height change and/or direction of growth) with that of either all-round light or no light (i.e. darkness) together with at least one control variable.

The bulk of the answers were in Level 1 as they contained some relevant points but did not satisfy the criteria described above. About 30% of students gave Level 2 answers and around 2% attained Level 3 in this question that differentiated across the ability range well.

A common error was to shine light onto the seedlings from different directions, possibly through variously-positioned holes cut into the sides and/or top of cardboard boxes. However, each of these still constituted light from one direction, so each group of seedlings was essentially treated in the same way (a Level 1 answer). Had an extra box with no holes been included, this would have given a control with the potential for entering a higher level. Some set up lamps at different distances hence giving different light intensities; this represented 'different light conditions' which made the answer potentially Level 2.

Many answers did not include any control variables, such as temperature and the amount of water given, the inclusion of which, along with a control set up, was a pre-requisite for entering Level 3.

Overall, attainment in this question was less than anticipated, especially considering that it tested knowledge and understanding of RPA 8 from the specification.

- (c) The key point of the phototropic response is that it enables a plant to absorb **more** light and thus perform photosynthesis **more** efficiently thereby producing **more** organic substances such as glucose.

Around 72% of students scored no marks in this question. A major cause of this was due to not giving a comparative answer. If a plant did not respond to the direction of light in this way, it would still receive some light and would still photosynthesise, but it would do so less well.

Other students included irrelevant details in their answers, attempting to explain the way redistribution of auxin could cause the phototropic effect rather than focusing on the advantages of it in terms of the plant's survival.